

INTRODUCTION TO ELECTRICAL ENGINEERING

[22ESC142/22ESC242]

QUESTION BANK

Sl. No	Questions	Marks	COs	BL
UNIT-1				
1	Explain the conventional and non-conventional energy sources	10	CO1	2,5
2	List Advantages and disadvantages of conventional energy sources	08	CO1	1,4
3	Advantages and disadvantages of non-conventional energy sources	08	CO1	1,4
4	With aid of single line diagram explain the general structure of electrical power system	08	CO1	2,5
5	Explain Solar power generation with neat block diagram	10	CO1	2,5
6	Explain Wind power generation with neat block diagram	10	CO1	2,5
7	Explain Hydel power generation with neat block diagram	10	CO1	2,5
8	Explain Nuclear power generation with neat block diagram	10	CO1	2,5
9	State and explain Ohm's law	06	CO1	2,5
10	State and explain Kirchhoff's law	06	CO1	2,5
11	Illustrate the equivalent resistance of two resistors connected in series is the sum of two resistance value.	06	CO1	2
12	Illustrate the equivalent resistance of two resistors connected in parallel is the ratio of product of these two resistances divided by the sum of those two-resistance value.	06	CO1	2
13	A circuit consists of two parallel resistors of 20 Ω and 30 Ω respectively, connected in series with 15 Ω . If the current through 15 Ω resistor is 3A, determine (i) current in 20 Ω and 30 Ω resistors, (ii) voltage across the whole circuit, (iii) the total power consumed in all resistances.	08	CO1	4
14	A current of 20A flows through 2 ammeters A & B in series. The potential difference across A is 0.2V & across B is 0.3V. Determine how the same current will divide between A&B when they are in parallel.	08	CO1	4
15	A circuit consisting of three resistors 12 Ω , 18 Ω , 36 Ω respectively joined in parallel is connected in series with a fourth resistance. The whole circuit is supplied at 60V and it is found that the power dissipated in the 12 Ω resistor is 36W. determine the fourth resistance and the total power dissipated in the group.	06	CO1	4
16	In a parallel resistive circuit resistors 8ohm, 40-ohm, 25 ohm and X ohm are connected in parallel. The current in 8-ohm resistor is 2.5A. The current entering the circuit is 4A. Determine (i) current in other resistors (ii) resistor X (iii) the equivalent resistance.	06	CO1	4
17	A 8-ohm resistor is in series with a parallel combination of two resistors 12-ohm and 6-ohm. If the current in the 6-ohm	08	CO1	4

	resistor is 5A, determine the total power dissipated in the circuit.			
	UNIT-2			
1	Sketch the sinusoidal AC waveform and define the following terms: (1) Time period (2) frequency (3) amplitude (4) phase (5) phase difference	06	CO1	1
2	Sketch the sinusoidal alternating current waveform and define the following : (i) Frequency (ii) Average value (iv) RMS value (v) Form Factor (vi) peak factor	06	CO1	1
3	Define the following: (1) Voltage (2) current (3) amplitude, (4) phase, (5) phase difference (6), average value (7), RMS value, (8) form factor (9) peak factor (10) power	10	CO1	1
4	Justify that a pure Inductor does not consume any power	06	CO1	5
5	Justify that a pure Capacitor does not consume any power	06	CO1	5
6	Illustrate with suitable circuit, phasor diagram and waveforms, that a pure inductor does not consume power	06	CO1	2
7	Illustrate with suitable circuit, phasor diagram and waveforms, that a pure resistor power is $P=VI$	06	CO1	2
8	Illustrate with suitable circuit, phasor diagram and waveforms, that a pure capacitor does not consume power	06	CO1	2
9	Justify that power consumed in an R-C series circuit is $VI\cos\phi$. Draw the waveform for voltage, current and power.	10	CO1	5
10	Justify that power consumed in an R-L series circuit is $VI\cos\phi$. Draw the waveform for voltage, current and power.	10	CO1	5
11	Justify that the power consumed in an RLC series circuit is $VI\cos\Phi$. Draw phasor for each case.	08	CO1	5
12	Draw the power triangle and define the following terms w.r.t AC circuits: (i) True power (ii) Apparent power (iii) Reactive power.	08	CO1	1
13	Define phase sequence and list out the advantages of three phase system as compared to single phase system.	06	CO1	1
14	Illustrate the relationship between the line and phase values of voltage and current in a three phase, delta connected circuit with relevant phasor diagram and circuit diagram	08	CO1	2
15	List the advantages of 3- ϕ systems over a single phase system	06	CO1	1
16	Illustrate the relationship between the line and phase values of voltage and current in a three phase, star connected circuit with relevant phasor diagram and circuit diagram	08	CO1	2
17	Elaborate the difference between balanced and unbalanced 3 Φ AC system.	06	CO1	6
18	The instantaneous values of the voltage across a two element series circuit and the current through it is given by v	06	CO2	5

	= $282 \sin(314t + 30^\circ)$ volts and $i = 7.05 \sin(314t - 30^\circ)$ amps. Determine a) Impedance b) Circuit elements c) power and d) power factor.			
19	A voltage of 200V is applied to a series circuit consisting of a resistor, an inductor and a capacitor. The respective voltages across these components are 170V, 150V, 100V and the current is 3A. Determine (i) the power factor (ii) resistance (iii) impedance (iv) inductive reactance and capacitive reactance.	08	CO2	5
20	A circuit consisting of a resistance of 10 ohms and an inductance of 12mH and a capacitance of $8\mu\text{F}$ all in series. A voltage of 150V at 50Hz is applied across the combination. Determine i) Impedance ii) current iii) power and iv) power factor	05	CO2	5
21	A coil of resistance 10Ω and inductance 0.1 H is connected in series with a $150\mu\text{F}$ capacitor across a 200V, 50 Hz supply. Determine the voltage across the coil and the capacitor respectively.	08	CO2	5
22	A series RLC circuit is composed of 100Ω resistance, 1H inductance and $5\mu\text{F}$ capacitance. A voltage $v(t)=1414.4\cos(377t)$ V is applied to the circuit. Determine the current and voltages of each element of the circuit.	08	CO2	5
23	The equation of an alternating current is given by $i=42.42\sin 628t$, determine its (i)maximum value (ii)frequency(iii)RMS value (iv)average value (v)Form factor.	06	CO2	5
24	An inductor coil is connected to a supply of 250V, at 50Hz and takes current of 4A. The coil dissipates 725W.Determine power factor, resistance and inductance of the coil.	06	CO2	5
25	Given $v=200\sin(377t)$ volts, $i=8\sin(377t-30^\circ)$ A for an AC circuit, determine (i)power factor (ii)active power (iii)reactive power(iv)apparent power.	08	CO2	5
26	The equation of an alternating current is given by $i=42.42\sin 628t$, determine its (i)maximum value (ii)frequency(iii)RMS value (iv)average value (v)Form factor.	06	CO2	5
27	A 50Ω resistor is connected in series with an inductive reactance of 70Ω to a 240V, 50Hz supply, determine (i) current through resistor and inductance, (ii) supply current, (iii) circuit phase angle. Draw the phasor diagram	06	CO2	5
28	A circuit consists of a resistance of 25Ω and a capacitance of $100\mu\text{F}$ connected in series. A supply of 200V at 50Hz is applied across the circuit. Determine the current, power factor and power consumed by circuit. Draw the vector diagram.	08	CO2	5

29	The three arms of a three phase load each comprise of an inductor of resistance of 25Ω , and of inductance $0.15H$, in series with a $120\mu F$ capacitor. The supply voltage is $415V$, $50Hz$. Determine the line current, and total power in watts, when the 3 arms are connected in delta	08	CO2	5
30	A balanced 3- ϕ star connected load of $150kW$ takes a leading current of $100A$, with a line voltage of $1100V$, $50Hz$. Determine the circuit constants of the load per phase	06	CO2	5
	A 3ϕ , $230V$, supply is given to balanced load which is Δ connected. Impedance in each phase of the load is $(8+j6)\Omega$. Determine the phase current and the total power consumed?	06	CO2	5
31	The 3 arms of a 3ϕ load each comprise of an inductor of resistance 20Ω and of inductance $0.10H$ in series with a $125\mu F$ capacitor. The supply voltage is $415V$, $50Hz$. Determine the line current and total power in watts, when the three arms are connected in star?	08	CO2	5
32	A balanced star connected load of $(8+6j)\Omega$ per phase is connected to a 3Φ $230V$ supply. Determine the line current, power factor, power, reactive volt-ampere and total VA.	10	CO2	5
33	A 3Φ delta connected load consumes a power of $60KW$ taking a lagging current of $200A$ at a line voltage of $400V$, $50Hz$. Determine the parameters of each phase. What would be power consumed, if the load were connected in star.	06	CO2	5
34	Three similar choking coils each having resistance 10Ω and reactance 10Ω are connected in star across a $440V, 3\Phi$ supply. Determine line current, power factor and power consumed	06	CO2	5
35	A 3Φ , Y-connected supply with a phase voltage of $230v$ is supplying a balanced delta load. The load draws $15KW$ at 0.8 p.f. lagging. Determine the line currents and the current in each phase of the load .Also find load impedance per phase	06	CO2	5
36	The 3 arm of a 3Φ load comprise of an inductor of resistance 20Ω and of inductance $0.10H$ in series with a $125\mu F$ capacitor. The supply voltage is $415V, 50Hz$. Evaluate the Phase current , line current and total power in watts , when the three arms connected in Delta.	08	CO2	5
37	A Y-connected load consists of 6Ω resistance and 8Ω inductive reactance in each phase. A supply of $440V$ at $50Hz$ is applied to the load. Determine the line current, P.f. and power consumed by the load.	06	CO2	5

UNIT-3: DC Machines				
Sl. No	Questions	Marks	COs	BL
1	State and explain Faraday's laws of electromagnetic induction.	06	3	5,2
2	Explain the working principle of a DC Machine as a generator and motor with suitable diagrams.	10	3	5,2
3	With neat sketch, explain the construction of a DC machine	10	3	5,2
4	Develop the expression for emf generated in a DC generator.	06	3	4,6
5	Explain the concept of back EMF and its significance	06	3	5,2
6	Explain the classification of DC motors	04	3	2
7	Sketch and explain the various characteristics of DC shunt motor and series motor and also mention their applications	10	3	5,2
8	Define Torque and develop the expression for armature torque developed in a DC motor	10	3	4,6
9	List the applications of the following motors. (i) DC shunt motor, (ii) DC series motor	04	3	4,6
10	An 8 pole generator has 400 armature conductors and has a useful flux per pole of 0.065Wb. Determine the e.m.f generated if it is lap connected and runs at 1000rpm and also evaluate the speed at which it is to be driven to produce the same e.m.f if it is wave wound.	10	4	5
11	A 6 pole DC generator has a lap connected armature with 480 conductors. The resistance of the armature circuit is 0.02Ω. With an output current of 500A from the armature, the terminal voltage is 230V when the machine is driven at 900rpm. Determine the useful flux per pole.	06	4	5
12	A four pole generator having wave-wound armature winding has 51 slots, each slot containing 20 conductors. Determine the voltage generated in the machine when driven at 1500 rpm assuming the flux per pole to be 7.0 mWb	10	4	5
13	An 8-pole d.c. generator has 500 armature conductors, and a useful flux of 0.05 Wb per pole. Determine the e.m.f. generated if it is lap-connected and runs at 1200 rpm and also	10	4	5

	determine the speed at which it is to be driven to produce the same e.m.f. if it is wave-wound.			
14	A 220V series motor is taking a current of 30A. Resistance of armature $0.5\ \Omega$, resistance of series field is $0.25\ \Omega$. Determine (i) voltage at the brushes (ii) back emf (iii) power wasted in armature (iv) power wasted in series field.	10	4	5
15	A 4 pole DC shunt motor takes 22A from 220V supply. The armature and the field resistances are $0.5\ \Omega$ and $100\ \Omega$ respectively. The armature is lap connected with 300 conductors. If the flux per pole is 20mWb, determine the speed and gross torque.	10	4	5
16	A 120V D.C. shunt motor has an armature resistance of $0.2\ \Omega$ and shunt field resistance of $60\ \Omega$. It turns at 1500rpm, when it takes full load current of 30A. Determine the speed of the motor while it is operating at half the full load.	10	4	5
17	A 200V, 4 pole, lap wound dc shunt motor has 600 conductors on its armature. The resistance of the armature winding is $0.5\ \Omega$ and shunt field winding is $200\ \Omega$. The motor takes a current of 21A, the flux per pole is 30mWb, Determine the speed and gross torque developed in the motor	10	4	5
18	A 440V dc shunt motor takes an armature current of 20A and runs at 500rpm. The armature resistance is $0.6\ \Omega$. If the flux is reduced by 30% and torque is increased by 40%. Determine the new values of armature current and speed.	10	4	5
19	A 20kW, 200V dc shunt motor has armature and field resistances of 0.05 ohm and 100 ohms respectively. Determine the total power developed by the armature when it delivers full output power.	10	4	5
20	A D.C series motor running with a speed of 1000rpm, while taking a current of 22amp from the supply. If the load is changed such that the current drawn by the motor is increased to 55amp, determine the speed of the motor on new load. The armature and series winding resistances are 0.3ohm and 0.4ohm respectively. Assume supply voltage as 250V.	10	4	5
21	A DC series motor connected to a connected to a 440V supply runs at 600rpm when taking a current of 50A. Determine the value of resistor which when inserted in series with the motor will reduce the speed to 400rpm, the gross torque being than half its previous value. Resistance of motor is $0.2\ \Omega$. Assume the flux to be proportional to the field current.	10	4	5
22	A series motor takes 20A at 400V and runs at 250rpm. The armature and field resistance are $0.6\ \Omega$ and $0.4\ \Omega$. Determine the applied voltage and current to run the motor at 350rpm, if the torque is required varies as the square of the speed.	10	4	5
23	A 4-pole DC shunt motor working on 250V takes a current of 2A when running light (i.e., no-load) at 1000rpm. Armature	10	4	5

	resistance and shunt field resistance are 0.2Ω and 250Ω . Determine (1) Back EMF generated (2) Back EMF and speed if the motor takes 51A at a certain load.			
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UNIT-4: Transformers				
Sl. No	Questions	Marks	COs	BL
1	Develop the relation between self-inductances, mutual inductance and co-efficient of coupling.	06	CO3	3
2	Define mutual inductance. Elaborate the expression for mutual inductance	06	CO3	1
3	Explain the concepts of statically induced EMF	06	CO3	2
4	Explain the working principle of operation of single-phase transformer.	06	CO3	2
5	Elaborate the main parts of transformer with neat sketch.	08	CO3	6
6	With neat sketch explain the construction of core and shell type transformer	10	CO3	5
7	Derive the EMF equation of transformer	08	CO3	1
8	List the applications of Transformers	04	CO3	1
9	Explain the various losses in a transformer and recommend measures to minimize them.	10	CO3	5
10	Define efficiency of the transformer. Develop efficiency expression at different loads.	08	CO3	6
11	Derive condition for maximum efficiency.	06	CO3	2
12	The maximum efficiency at full load and unity power factor of a single phase 25 KVA, 500/1000V, transformer is 98%, Determine its efficiency at $3/4^{\text{th}}$ full load	08	CO4	5
13	A 600 KVA transformer has an efficiency of 92% at full load, upf and at half load, 0.9 Pf. Determine its efficiency at 75% of full load and 0.9 p.f.	08	CO4	5
14	A 40 KVA single phase transformer has core loss of 450W and full load copper loss of 850W. if the power factor of the load is 0.8 determine Full load efficiency, Maximum efficiency at UPF.	08	CO4	5

15	A 600 KVA, transformer has efficiency at 92% both at full load and half full load upf, determine its efficiency at 75%fullload, 0.9 pf.	06	CO4	5
16	A transformer is rated at 100kva. At full load its copper loss is 1200W and its iron loss is 960 W determine efficiency at full load, upf. Efficiency at half load, 0.8pf. The load KVA at which maximum efficiency occur.	06	CO4	5
17	A 125KVA transformer has a primary voltage of 2000V, 50Hz, primary turns are 182 and the secondary turns 40. Neglecting the losses, evaluate: (i) no load secondary EMF, (ii) full load primary and secondary currents, (iii) flux in the core.	06	CO4	5
18	A 250KVA, 11000/415 V, 50Hz 1 Φ transformer has 80 turns on the secondary. Determine (i) The rated primary and secondary currents (ii) The number of primary turns (iii) The maximum value of flux (iv) voltage induced per turn.	08	CO4	5
19	The primary winding of a transformer is connected to a 240V, 50Hz supply. The secondary winding has 1500 turns. If the maximum value of core flux is 0.00207 wb. Determine (i) The secondary induced EMF (ii) Number of turns in the primary (iii) Core area of cross section if the flux density has a maximum value of 0.465 Tesla.	06	CO4	5
	UNIT-4: 3-Phase Induction Motors			
Sl. No	Questions	Marks	COs	BL
1	Discuss the important features of squirrel cage and phase wound rotor constructions of an induction motor with neat diagram.	08	CO3	6
2	Explain the working principle of three phase induction motor	06	CO3	2
3	Elaborate slip of the induction motor and under what circumstances the slip is i) unity ii) zero	08	CO3	6
4	Explain synchronous speed, slip speed and motor speed in case of three phase induction motor. Justify slip is never zero in an induction motor.	08	CO3	5
5	List the applications of three phase induction motor.	06	CO3	1
6	Develop the expression for the slip and frequency of the rotor currents.	06	CO3	6
7	Discuss the advantages, disadvantages and applications of three phase induction motors.	08	CO3	2
8	Discuss the concept of rotating magnetic field with neat phasors.	10	CO3	6

9	A 6 pole IM is supplied by a 10-pole alternator which is driven at 600 RPM. If the motor is running at 970 rpm, determine the percentage slip.	06	CO4	5
10	A 12 pole, 3 phase alternator is coupled to an engine running at 500 RPM It supplies an IM, which has full load speed of 1440r.p.m determine the percentage slip and the number of poles the motor.	06	CO4	5
11	A –phase, 6 pole, 50Hz IM has a slip of 1% at no-load, and 3% at full load. Determine: i) Synchronous speed; ii) no-load speed; iii) full-load speed; iv) frequency of the rotor current at stand still; v) frequency of the rotor current at full load.	10	CO4	5
12	A 3-phase, 8 pole, 50Hz IM has a slip of 1% at no-load, and 3% at full load. Determine; I) synchronous speed, ii) No-load speed, iii) Full load speed, iv) Frequency of rotor at standstill	08	CO4	5
13	The rotor induced voltage of a 3-phase, 4 –pole squirrel cage induction motor fed by a salient pole alternator is observed to make 1.5 alterations per second. The star connected alternator with 592, full pitched armature conductors in series per phase with distribution factor of 0.966 develops a line voltage of 6600 volts when the flux per pole is 60mWb. Determine the speed of the IM.	08	CO4	5
14	A 12 pole, 3 phase alternator is coupled to an engine running at 500r.p.m It supplies an IM, which has full load speed of 1440r.p.m determine the percentage slip and the number of poles the motor.	06	CO4	5
15	A 3phase 50Hz, 6 pole induction motor has full load percentage slip of 3%. Evaluate for synchronous speed and actual speed.	06	CO4	5
16	A three phase 50Hz 6-pole induction motor has a full load percentage slip of 3%. Determine (i) Synchronous speed (ii) Actual speed (iii) rotor frequency.	06	CO4	5
17	8-pole alternators run at 750 rpm supplies power to induction motor. The frequency of rotor is 1.5Hz. Determine the speed of motor and slip	06	CO4	5

UNIT-5				
Sl. No	Questions	Marks	COs	BL
1	Explain the conduit type of domestic wiring with a neat sketch	06	CO5	5
2	Explain the requirements of domestic wiring	04	CO5	2
3	With a neat sketch explain domestic wiring of a house	08	CO5	2

4	Discuss two ways of staircase wiring with neat sketch and on-off table.	10	CO5	6
5	Explain two-way control of lamp	06	CO5	2
6	Explain three-way control of lamp	06	CO5	2
7	Elaborate electricity billing and tariff	08	CO5	6
8	With an example explain power requirement calculation for domestic applications.	06	CO5	5
9	With an example explain electricity utility billing	06	CO5	5
10	Explain difference between KWH and KVAH	04	CO5	2
11	Explain Two-part tariff with an example	06	CO5	2
12	With a aid of neat sketch explain FUSE	08	CO5	2
13	With a neat sketch explain MCB	10	CO5	5
14	List the Merits and Demerits of FUSE and MCB	10	CO5	1
15	Explain the necessity and the operation of miniature circuit breaker.	10	CO5	2
16	Define Electric shock. List out safety measures against electric shock.	08	CO5	1
17	Discuss Safety Precautions to avoid shock	10	CO5	6
18	Explain the necessity of earthing	04	CO5	2
19	With a neat sketch explain types of earthing	10	CO5	5
20	Explain pipe earthing with a neat sketch	08	CO5	5
21	Explain plate earthing with a neat diagram	08	CO5	5
22	If a 40-watt lamp is turned on for one hour, determine joules of electrical energy converted by the lamp	04	CO5	5
23	Evaluate the heat produced by an electric iron, which has a resistance of 30 ohms and takes a current of 3 amperes when it is switched on for 15 seconds.	06	CO5	5
24	A consumer consumes 1000 watts load per hour daily for one month. Determine the Total Energy bill of the consumer if per unit rate is 9 INR [Take 1 month = 30 Days].	06	CO5	5
25	A Consumer has a maximum demand of 50KW and the energy consumed per month is 500units, determine the monthly electrical charges of the consumer. Assuming a = Rs.2/KW and b = 50 paise/KWh.	04	CO5	5
26	Determine the units of electricity consumed in the month of November from the following details. One 60W60W bulb is used for 55 hours daily. One 100W100W bulb is used for 33 hours daily. One 1kW1kW electric heater is used for 22 hours daily.	06	CO5	5